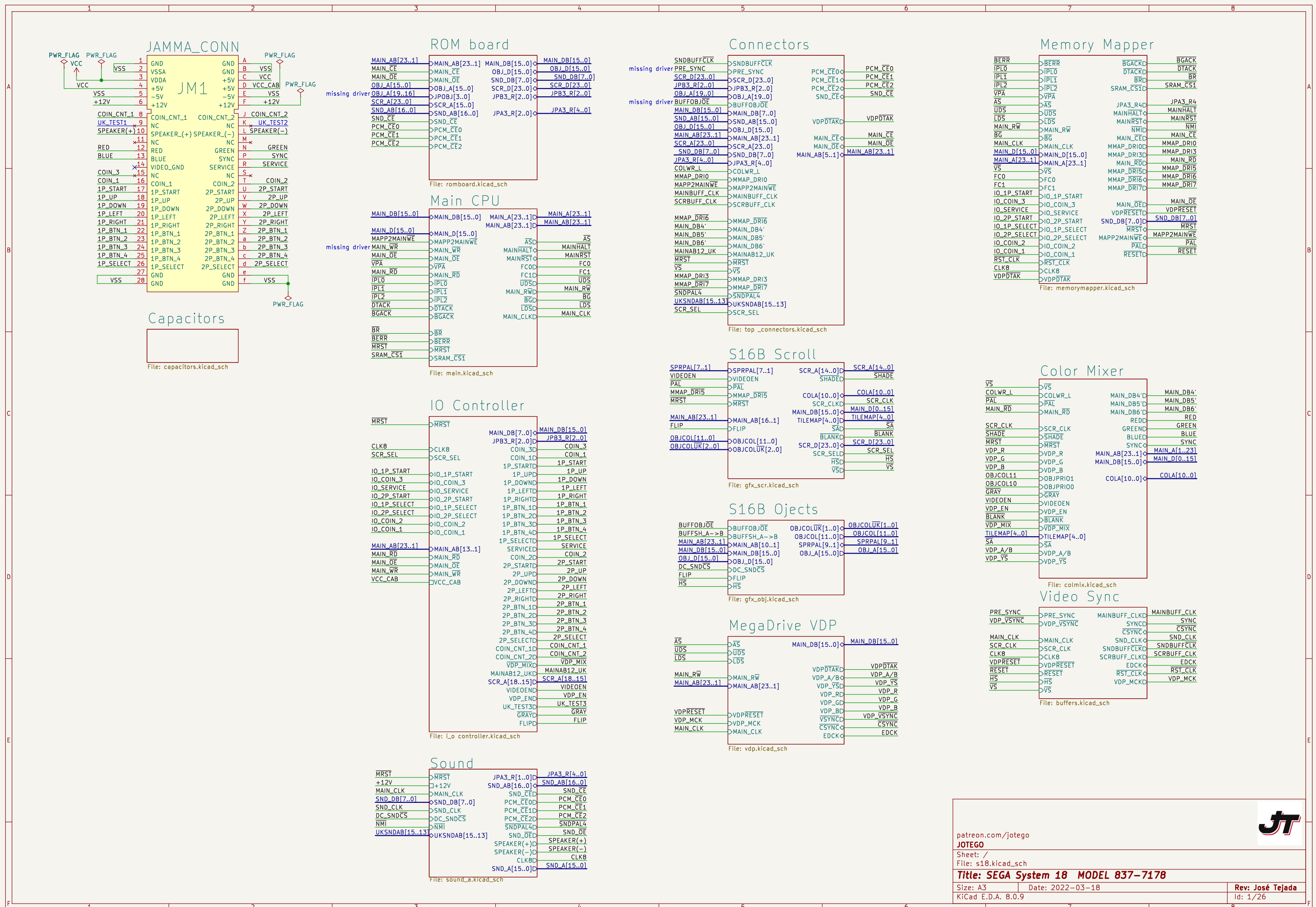
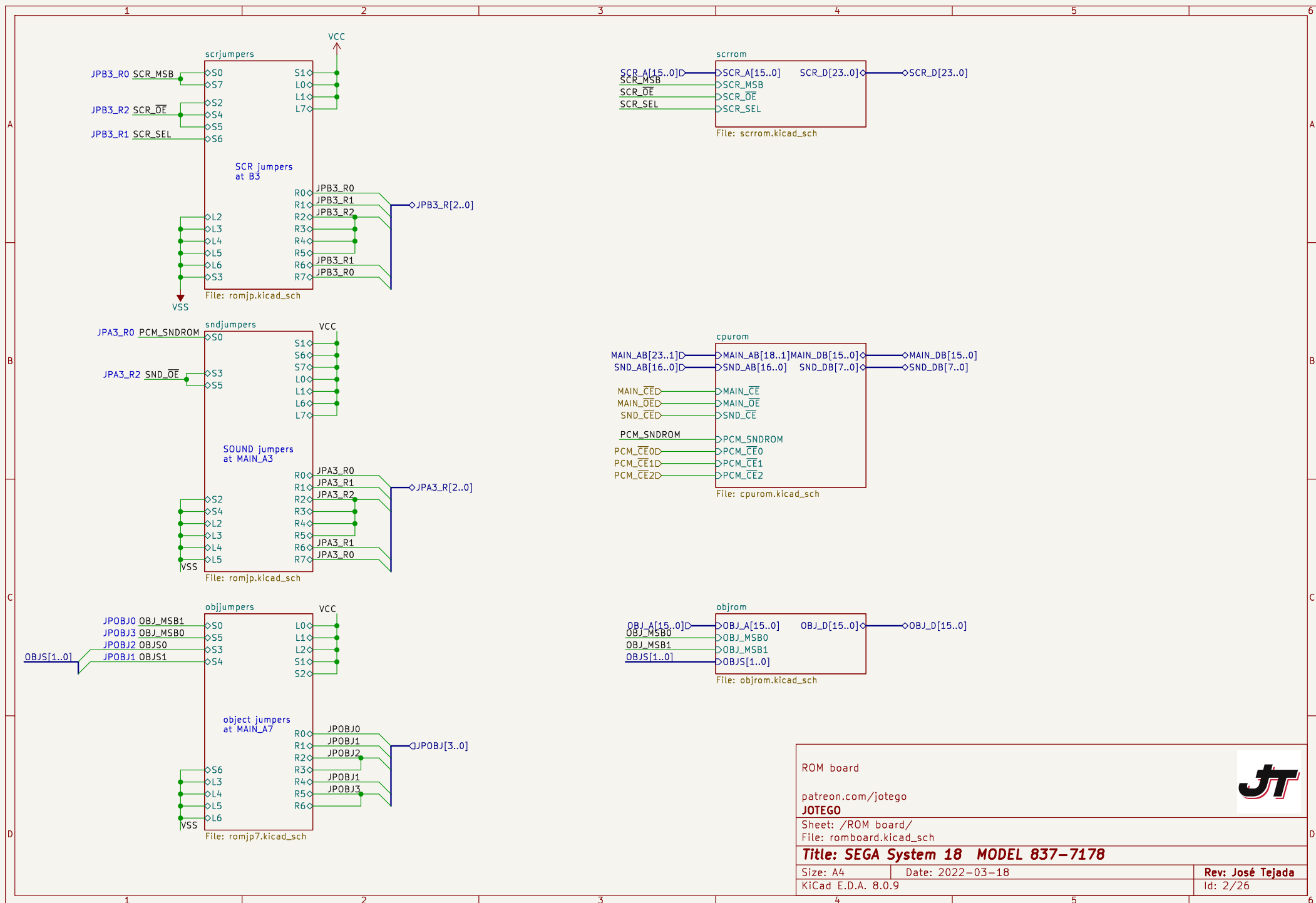






ROM board

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JOTEGO

Sheet: /ROM board/

File: romboard.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

Date: 2022-03-18

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Rev: José Tejada

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Sheet: /ROM board/scrjumpers/

File: romjp.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

Date: 2022-03-18

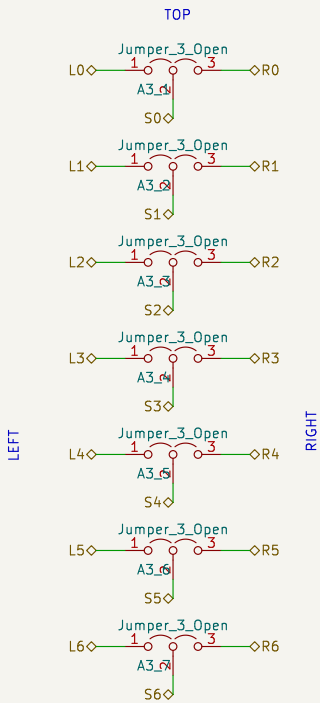
Rev: José Tejada

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Id: 3/26







BOTTOM

Seven jumpers

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JOTEGO

Sheet: /ROM board/objjumpers/

File: romjp7.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

Date: 2022-03-18

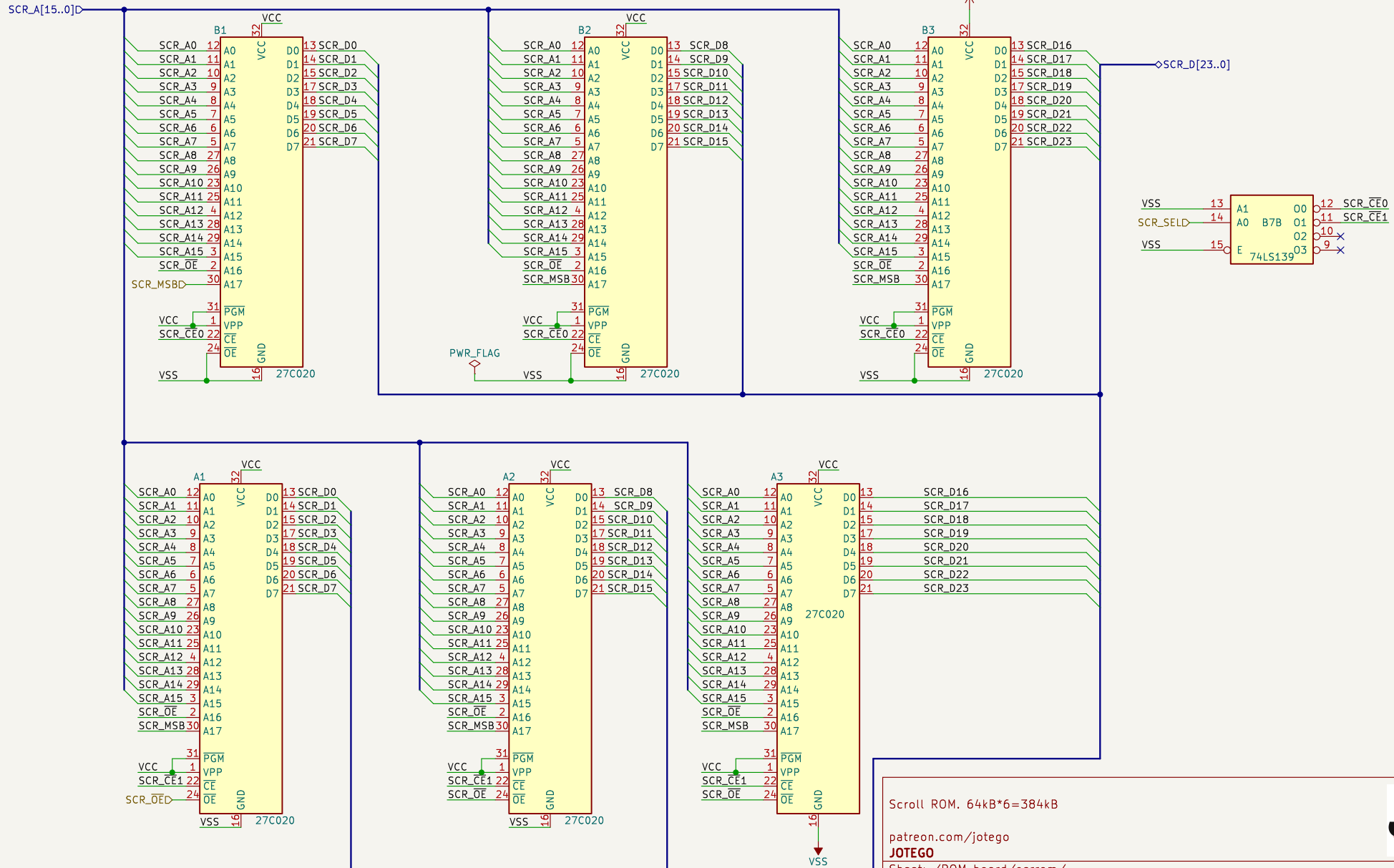
Rev: José Tejada

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Id: 5/26



SCROLL ROM



Scroll ROM. 64kB*6=384kB

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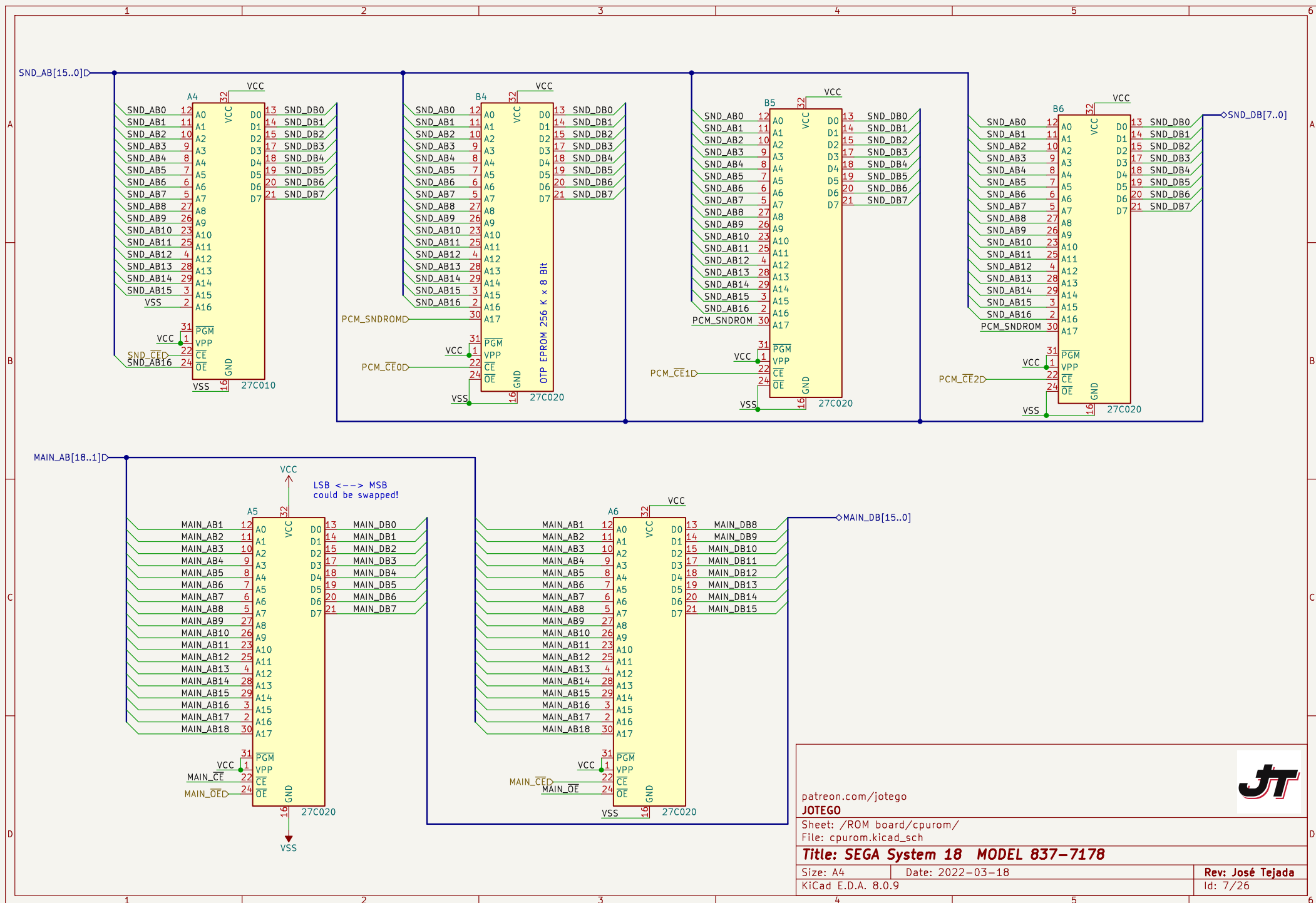
Sheet: /ROM board/scrrom/
File: scrrom.kicad_sch

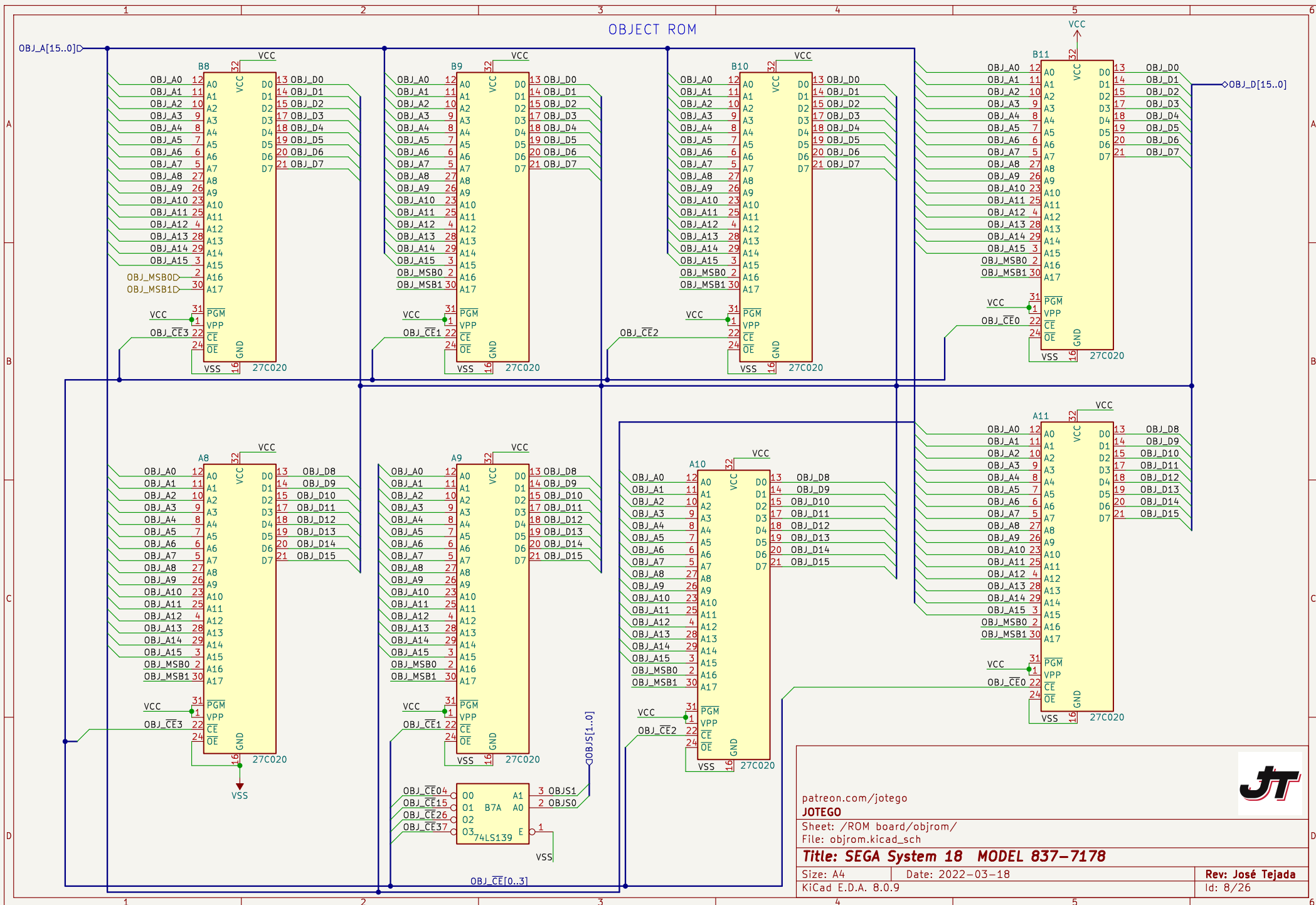
Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18
KiCad E.D.A. 8.0.9

Rev: José Tejada
Id: 6/26







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Sheet: /ROM board/objrom/

File: objrom.kicad_sch

Title: SEGA System 18 MODEL 837-7178

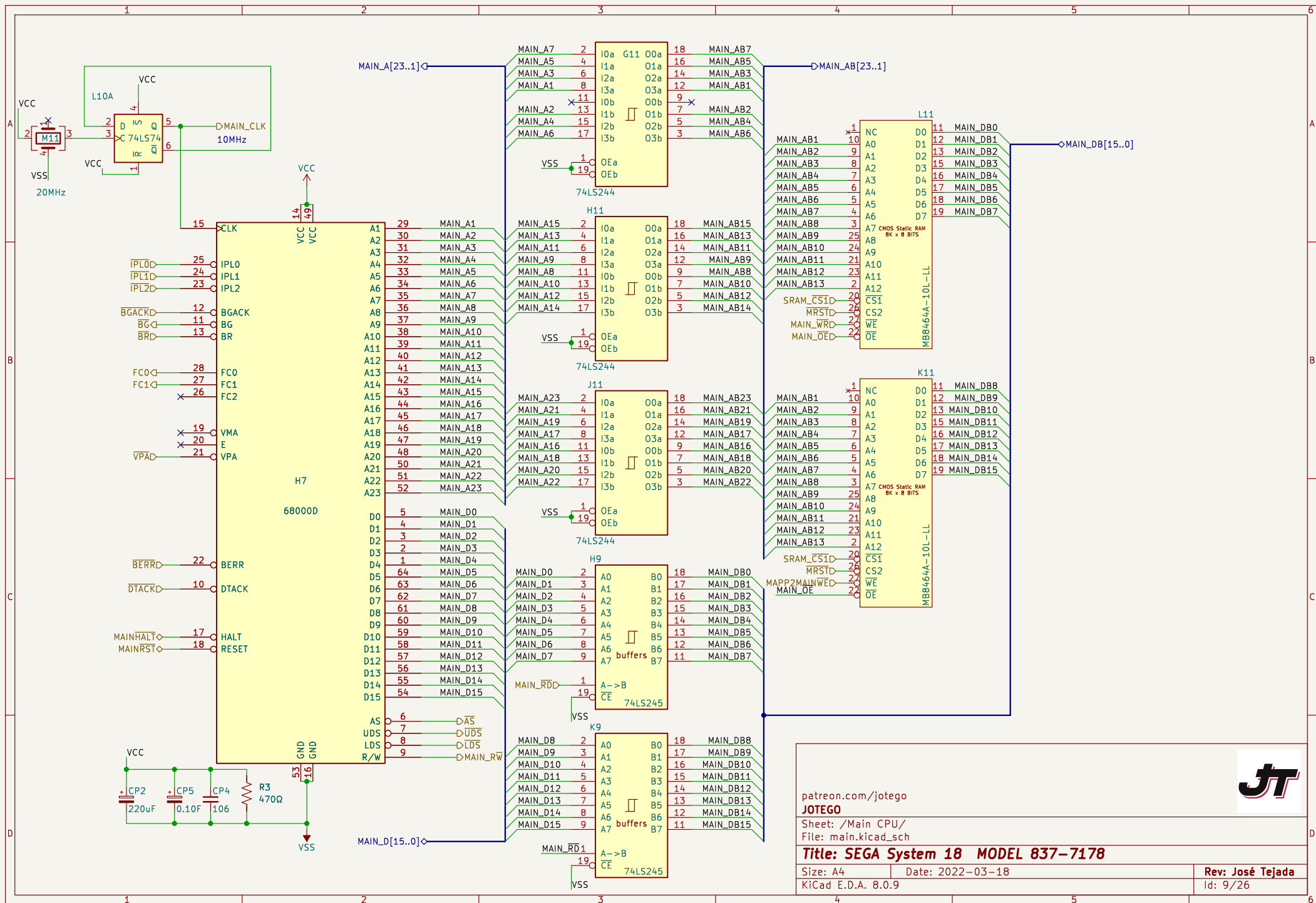
Size: A4 Date: 2022-03-18

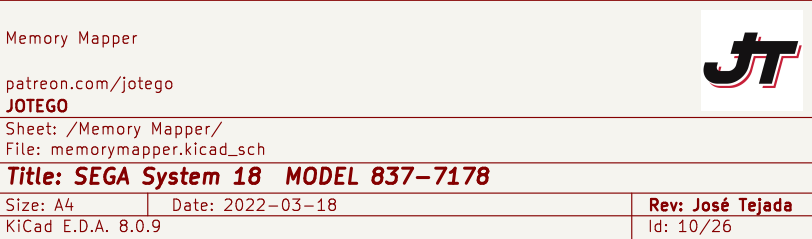
KiCad E.D.A. 8.0.9

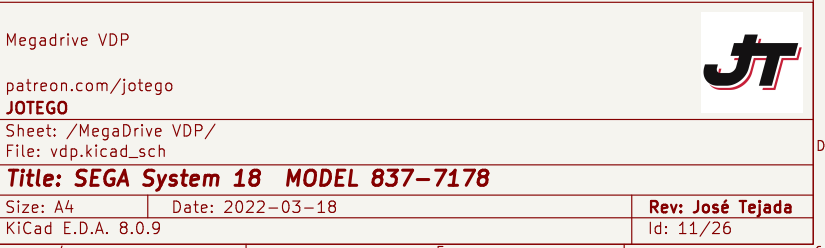
Rev: José Tejada

Id: 8/26



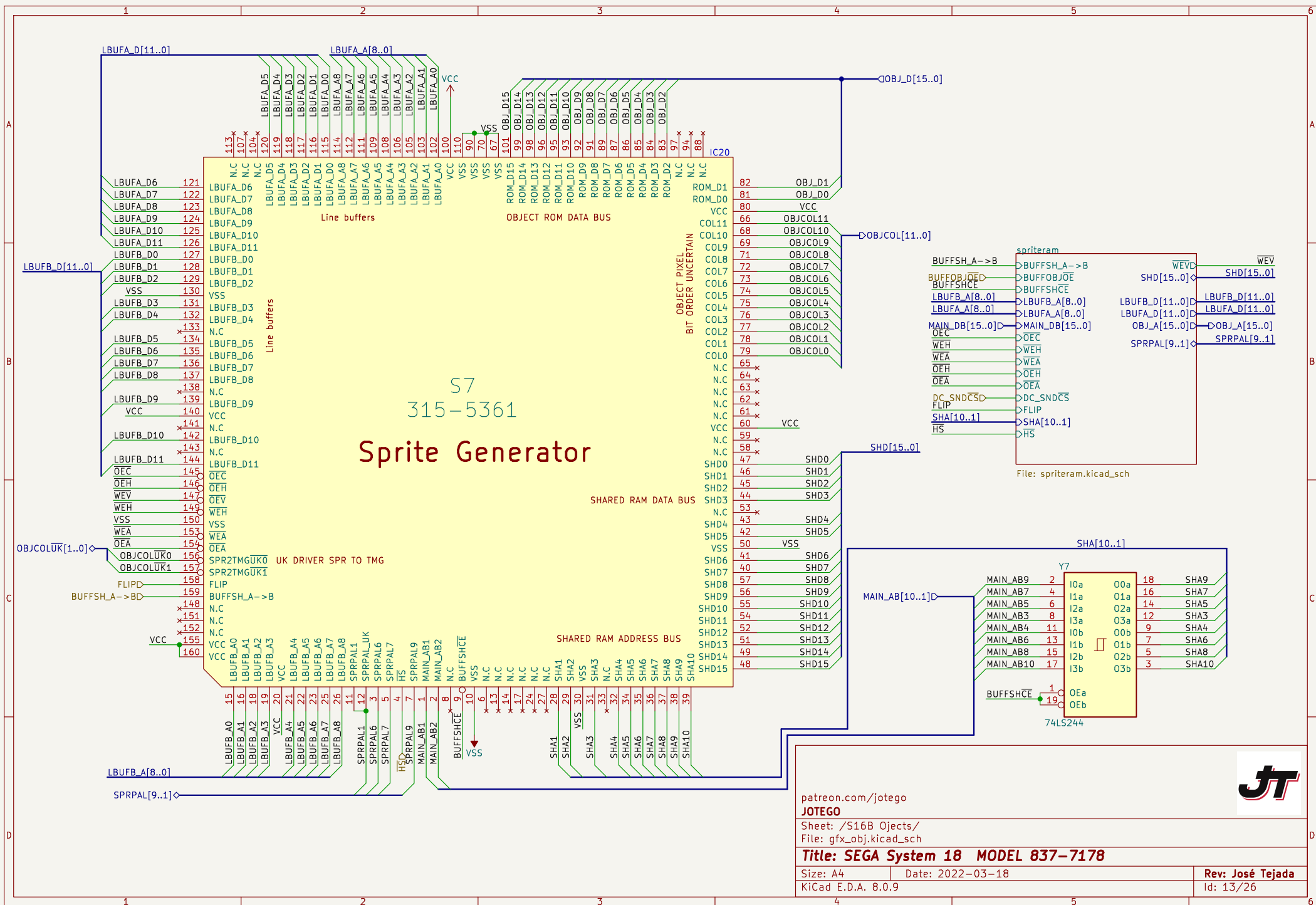


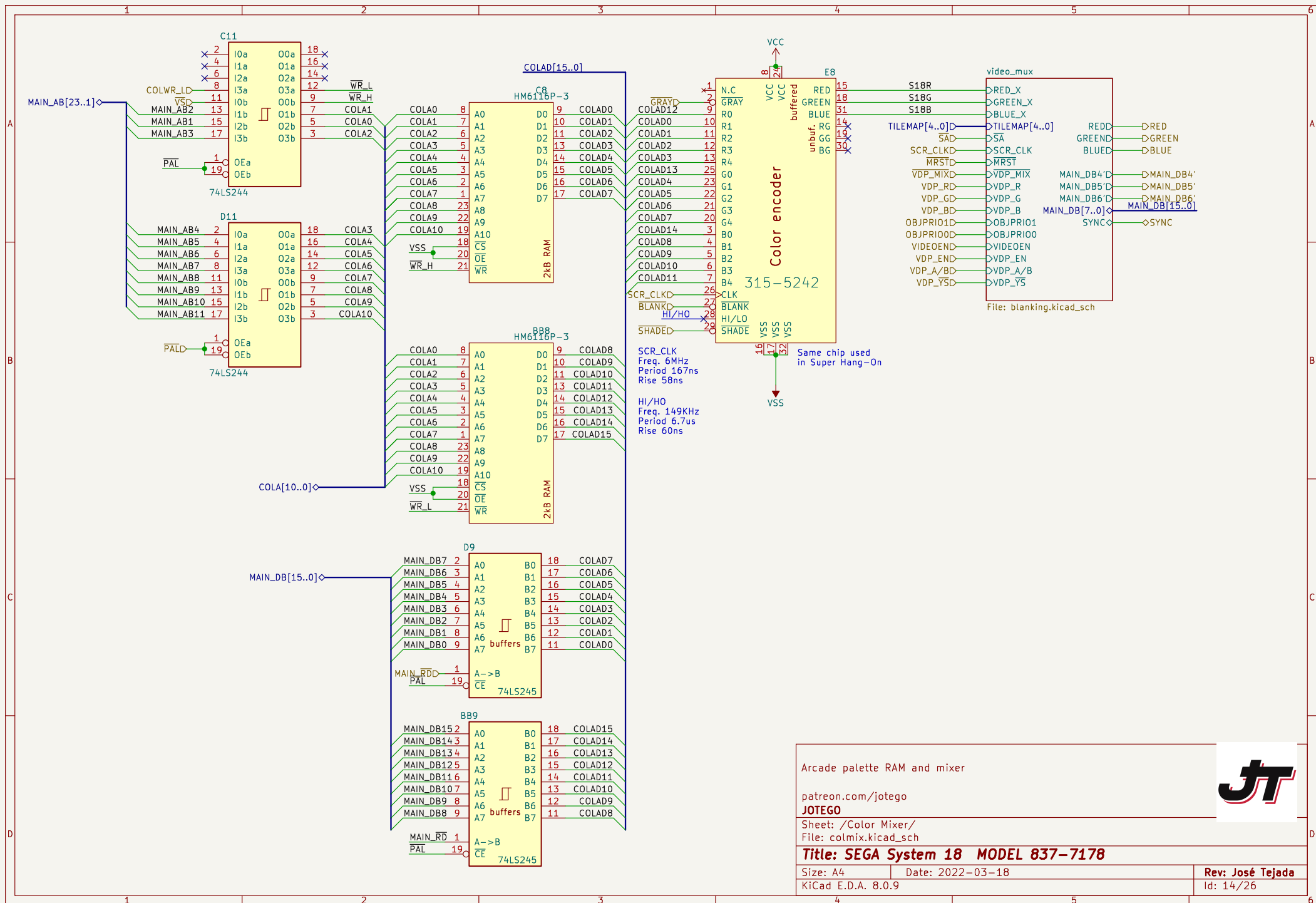






Id: 12/26





Arcade palette RAM and mixer

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JOTEGO

Sheet: /Color Mixer/

File: colmix.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

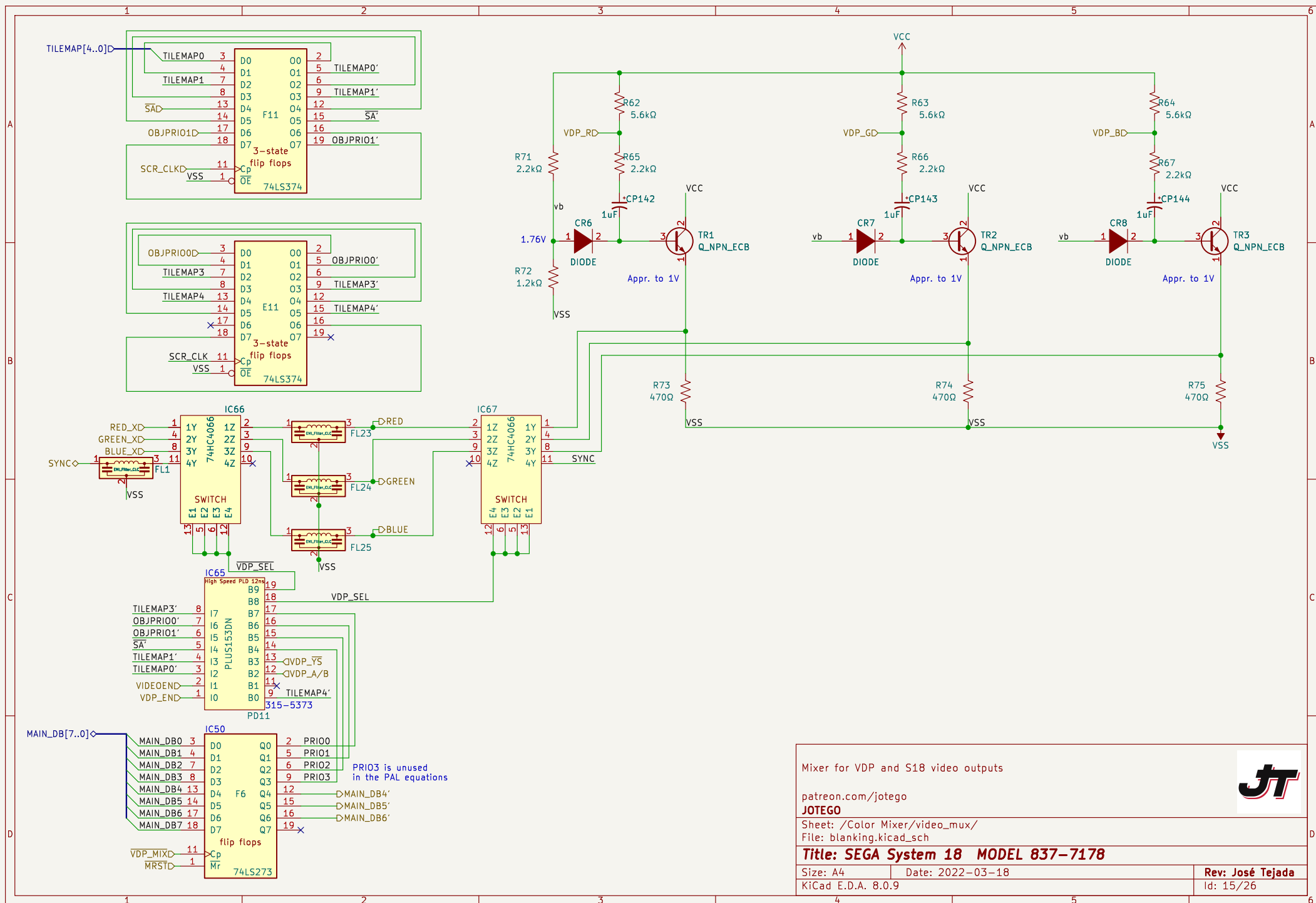
Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 14/26





Mixer for VDP and S18 video outputs

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JOTEGO

Sheet: /Color Mixer/video_mux/

File: blanking.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 15/26





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Sheet: /Capacitors/
 File: capacitors.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

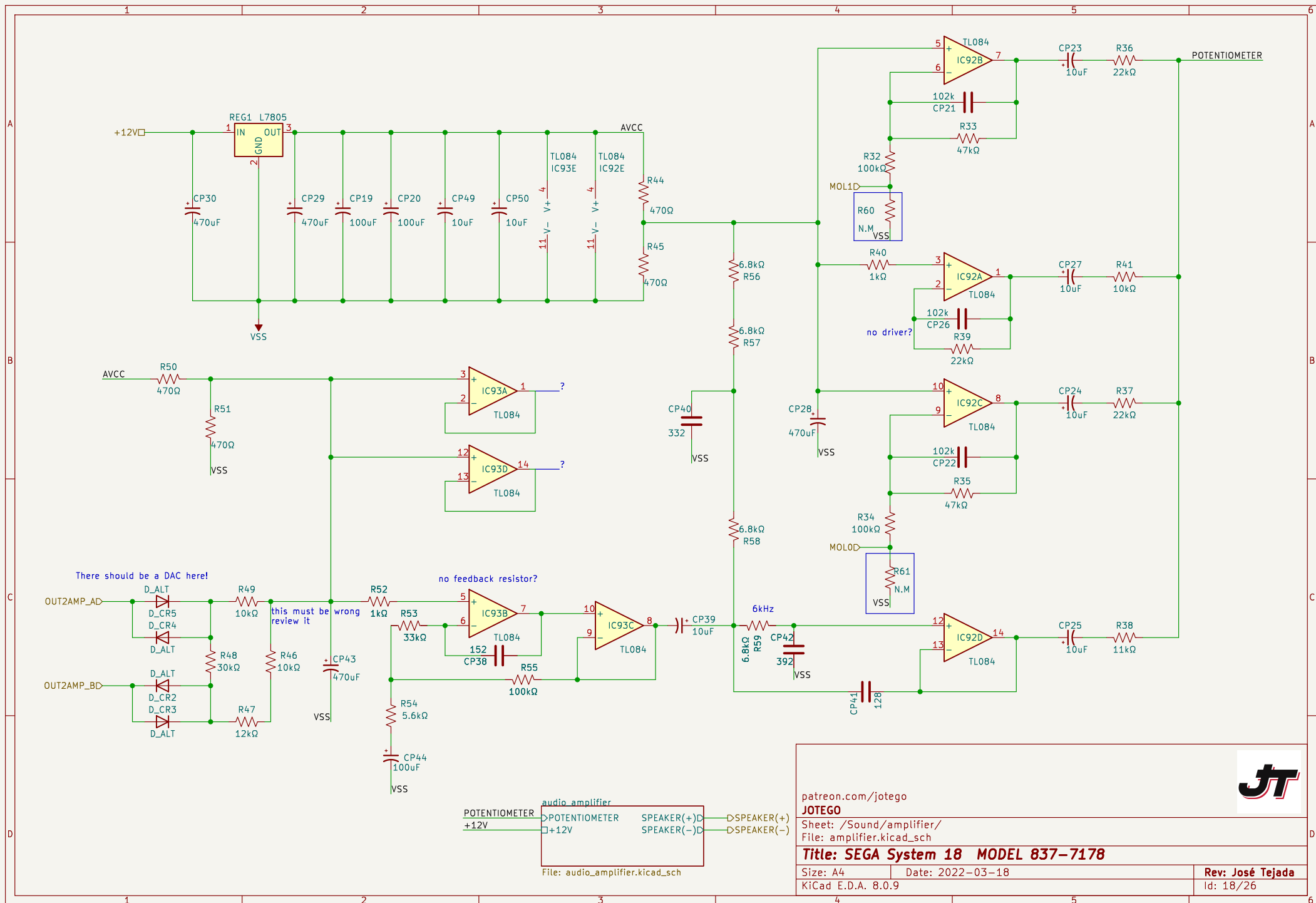
Rev: José Tejada

Id: 16/26





Rev: José Tejada
Id: 17/26



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JOTEGO

Sheet: /Sound/amplifier/

File: amplifier.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 18/26



MOTHER AND DAUGHTER PCB CONNECTORS

Connector CN1:

Signal	Pin	Signal	Pin
SCR_D[23..0]D	a1	SCR_D1	b1
	a2	SCR_D3	b2
	a3	SCR_D5	b3
	a4	SCR_D7	b4
	a5	SCR_D9	b5
	a6	SCR_D11	b6
	a7	SCR_D13	b7
	a8	SCR_D15	b8
	a9	SCR_D17	b9
	a10	SCR_D19	b10
	a11	SCR_D21	b11
	a12	SCR_D23	b12
SCR_A[23..0]D	a13	SCR_A14	b13
	a14	SCR_A12	b14
	a15	SCR_A10	b15
	a16	SCR_A8	b16
	a17	SCR_A6	b17
	a18	SCR_A4	b18
	a19	SCR_A2	b19
	a20	SCR_A0	b20
	a21	JPB3_R0	b21
	a22	JPB3_R1	b22
JPB3_R[2..0]	a23		b23
	a24		b24
MAINBUFF_CLK	a25		b25
VSS	a26	VSS	b26
	a27	VSS	b27
	a28	VSS	b28
VCC	a29	VCC	b29
	a30	VCC	b30

Connector CN2:

Signal	Pin	Signal	Pin
OBJ_D[15..0]D	a1	OBJ_D1	b1
	a2	OBJ_D3	b2
	a3	OBJ_D5	b3
	a4	OBJ_D7	b4
	a5	OBJ_D9	b5
	a6	OBJ_D11	b6
	a7	OBJ_D13	b7
	a8	OBJ_D15	b8
	a9	OBJ_A16	b9
	a10	OBJ_A17	b10
	a11	OBJ_A1	b11
	a12	OBJ_A3	b12
	a13	OBJ_A5	b13
	a14	OBJ_A7	b14
	a15	OBJ_A9	b15
	a16	OBJ_A11	b16
	a17	OBJ_A13	b17
	a18	OBJ_A15	b18
	a19	OBJ_A19	b19
	a20	OBJ_A18	b20
	a21	OBJ_A17	b21
	a22	OBJ_A16	b22
	a23	OBJ_A15	b23
	a24	OBJ_A14	b24
	a25	OBJ_A13	b25
VSS	a26	VSS	b26
	a27	VSS	b27
	a28	VSS	b28
VCC	a29	VCC	b29
	a30	VCC	b30

Connector CN3:

Signal	Pin	Signal	Pin
MAIN_AB[23..1]D	a1	MAIN_AB2	b1
	a2	MAIN_AB4	b2
	a3	MAIN_AB6	b3
	a4	MAIN_AB8	b4
	a5	MAIN_AB10	b5
	a6	MAIN_AB12	b6
	a7	MAIN_AB14	b7
	a8	MAIN_AB16	b8
	a9	MAIN_AB18	b9
	a10	MAIN_AB20	b10
	a11	MAIN_AB22	b11
	a12	SND_AB0	b12
	a13	SND_AB2	b13
	a14	SND_AB4	b14
	a15	SND_AB6	b15
	a16	SND_AB8	b16
	a17	SND_AB10	b17
	a18	SND_AB12	b18
	a19	UKSNDAB15	b19
	a20	UKSNDAB14	b20
	a21	SND_CE	b21
	a22	PCM_CE0	b22
	a23	PCM_CE1	b23
	a24	SCRBUFF_CLKD	b24
	a25	MAIN_DB5	b25
	a26	MAIN_DB6	b26
	a27	VSS	b27
	a28	VSS	b28
VCC	a29	VCC	b29
	a30	VCC	b30

Connector CN4:

Signal	Pin	Signal	Pin
VCC	a25	VCC	b25
VSS	a26	VSS	b26
COLWR_LD	a27		b27
JPA3_R[4..0]D	a28		b28
	a29		b29
	a30		b30
MAIN_CE	a31		b31
MAIN_OE	a32		b32
SND_DB[7..0]D	a33		b33
	a34		b34
	a35		b35
	a36		b36
	a37		b37
	a38		b38
	a39		b39
	a40		b40
	a41		b41
	a42		b42
	a43		b43
	a44		b44
	a45		b45
	a46		b46
	a47		b47
	a48		b48
	a49		b49

Used as ADC
inputs from gun
in Laser Ghost
on board:
837-7536
171-5993A

JOTEGO

File: top_connectors.kicad_sch

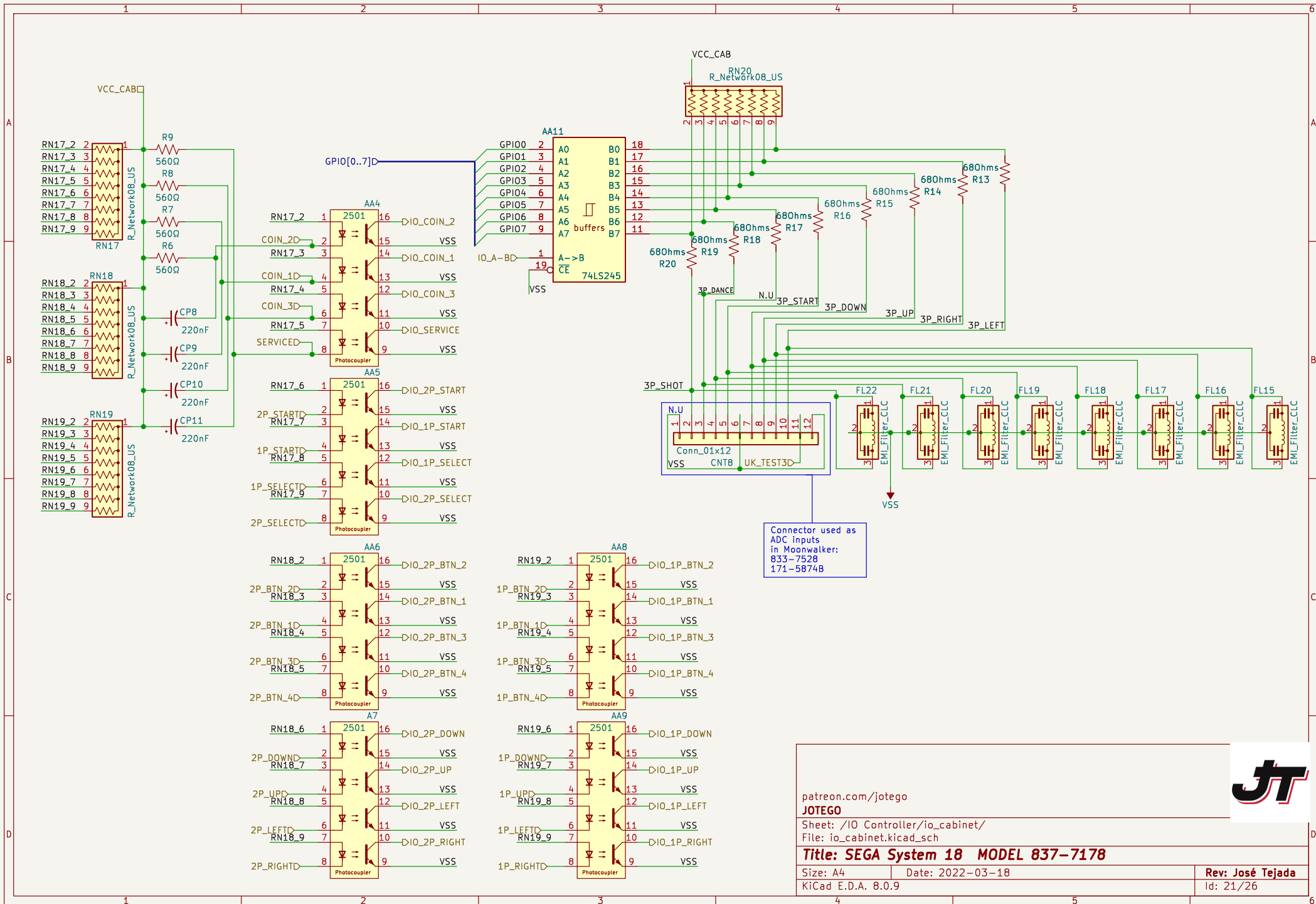
Title: SEGA System 18 MODEL 837-7178

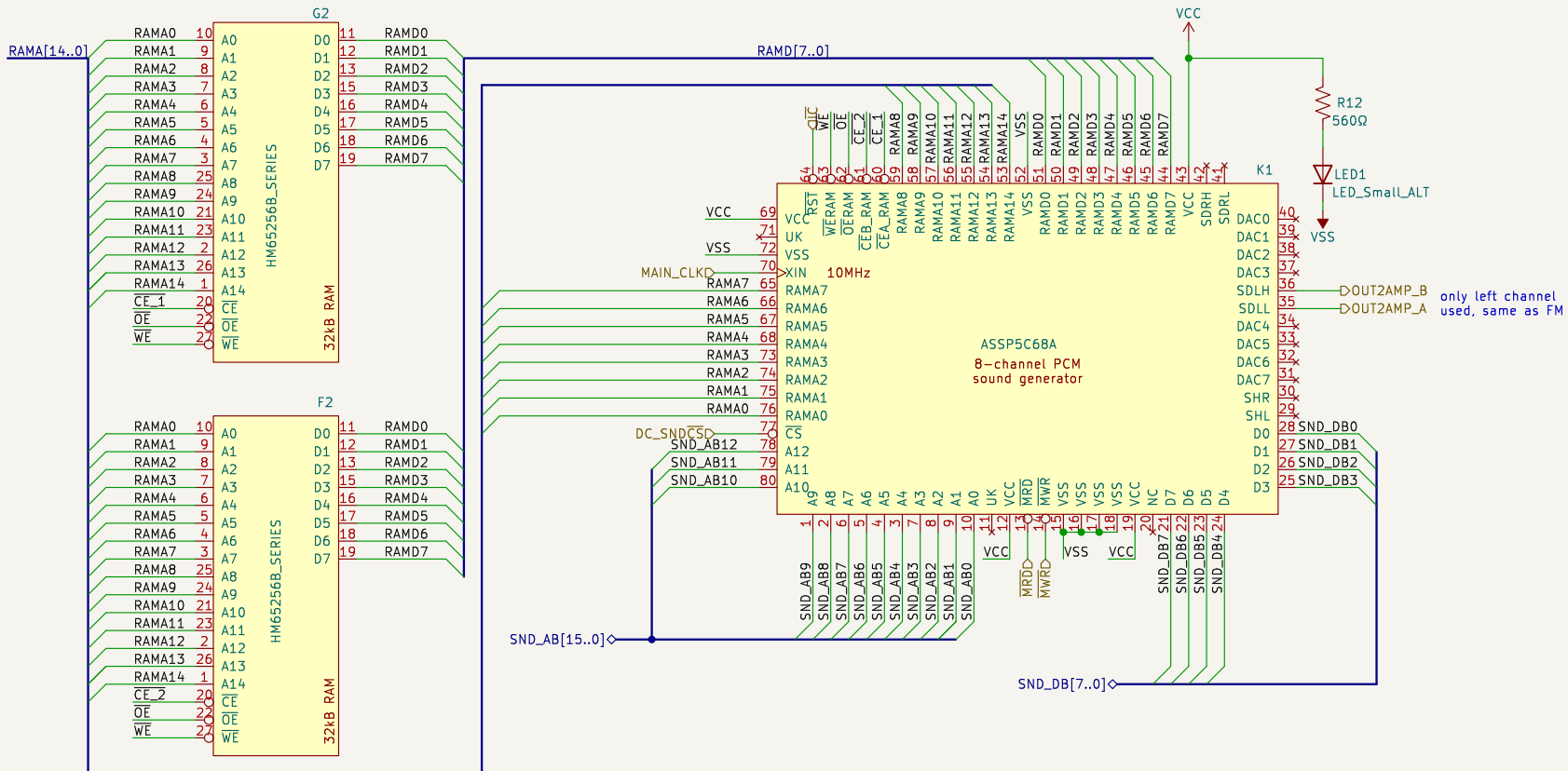
Size: A4
KiCad E.D.A. 8.0.9

Id: 19/26









PCM sound generator

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Sheet: /Sound/PCM/

File: comm_snd.kicad_sch

Title: SEGA System 18 MODEL 837-7178

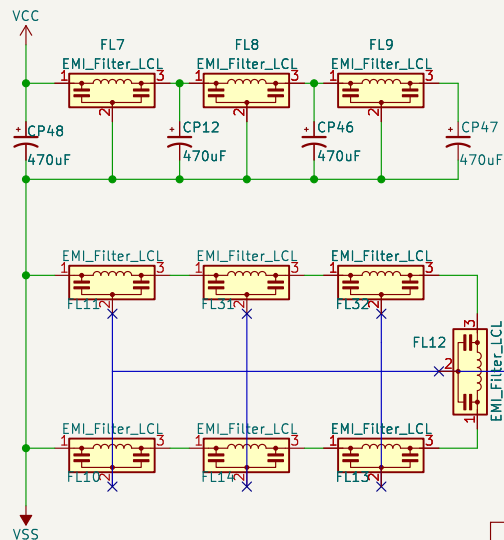
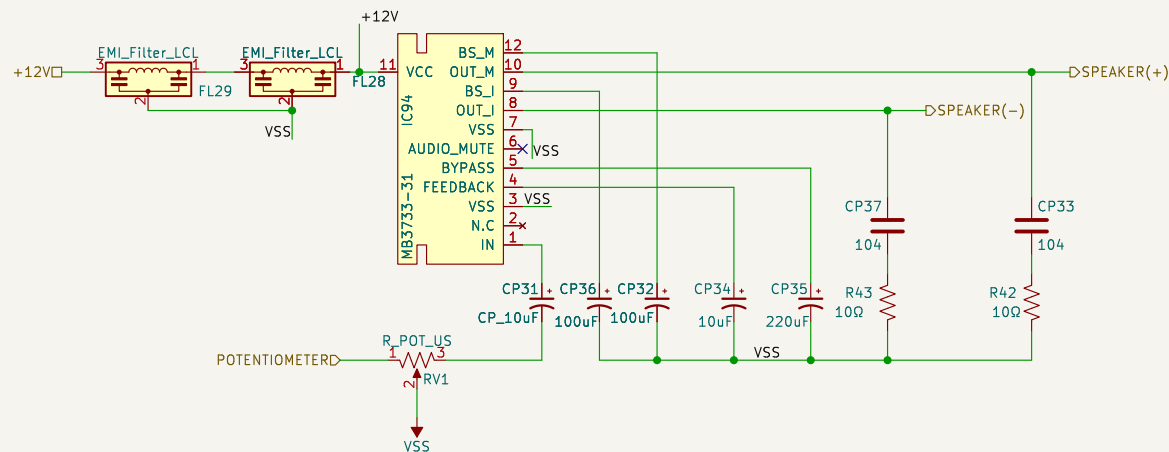
Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 22/26





Leaded EMIFL
Three-terminal Capacitor type
DSS6NE52A222p
2200pF 80/-20%
100Vdc 6A -25 to +85°C

A kind of a track is observed between the two PCB layers, as if it were another layer, but there is no certainty about that. The track seems to be used to delimit or isolate the EMI (ElectroMagnetic Interference) filters. They may have wanted to isolate the noise between digital and analog devices.

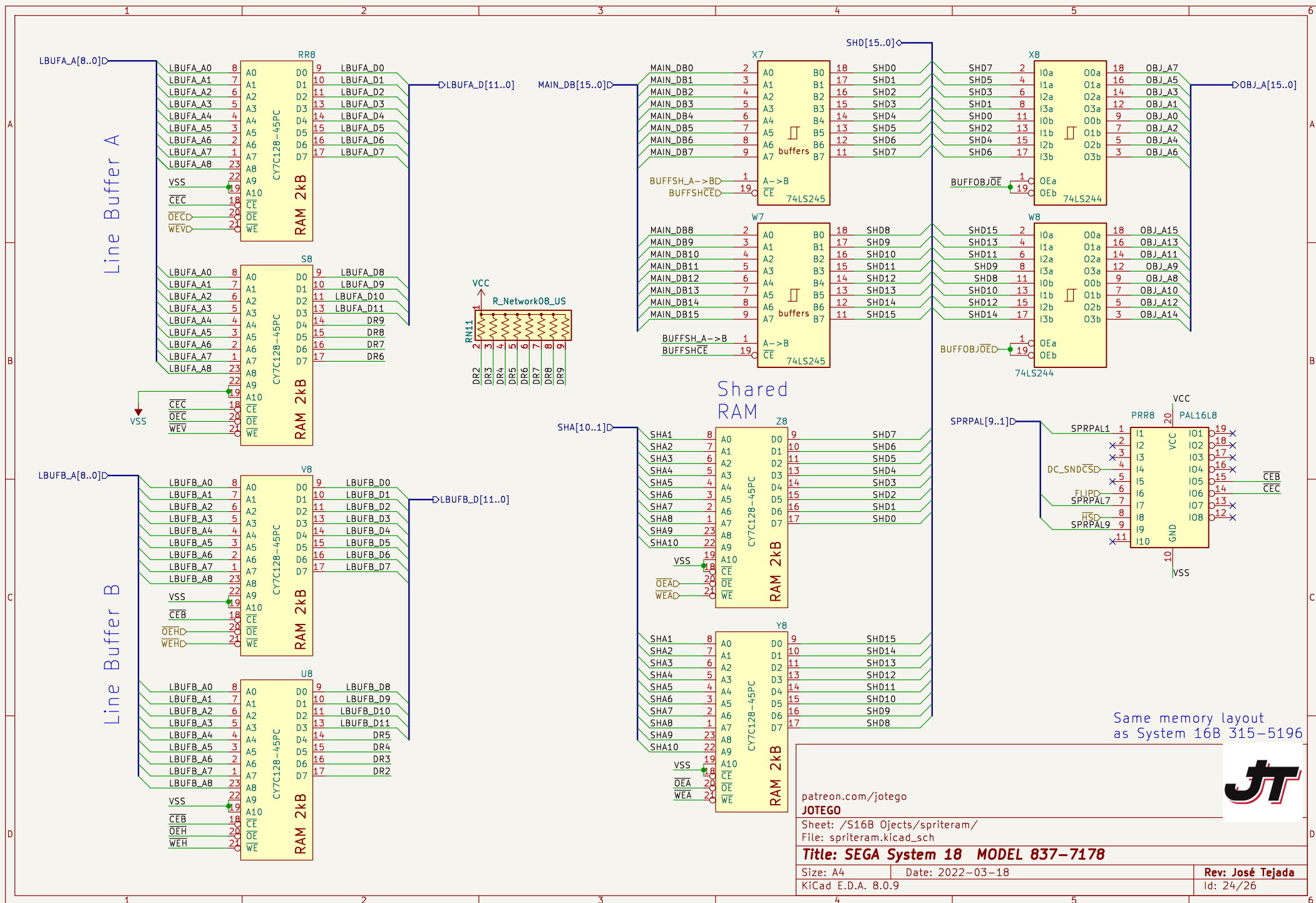
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JOTEGO

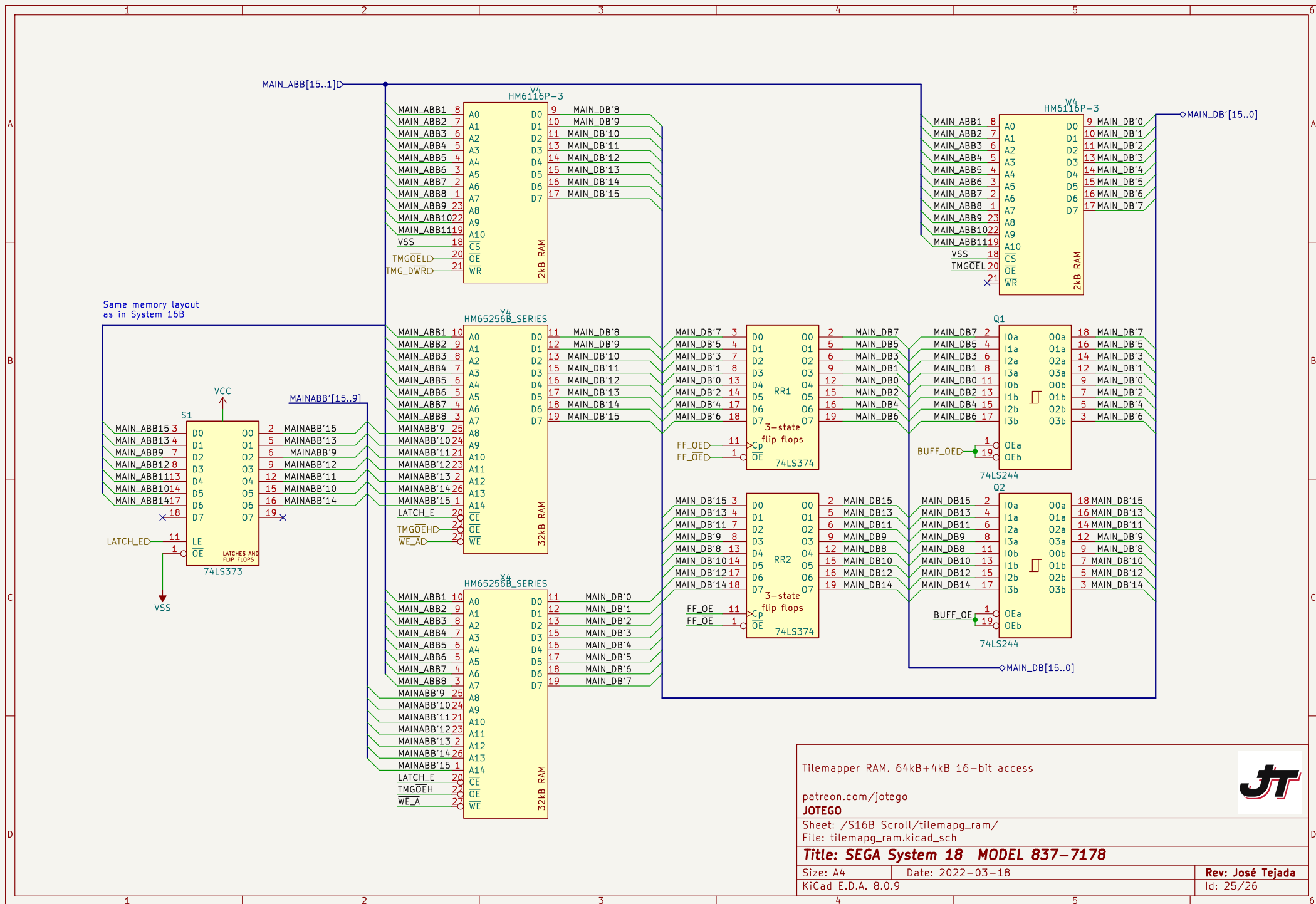
Sheet: /Sound/amplifier/audio_amplifier/
File: audio_amplifier.kicad_sch

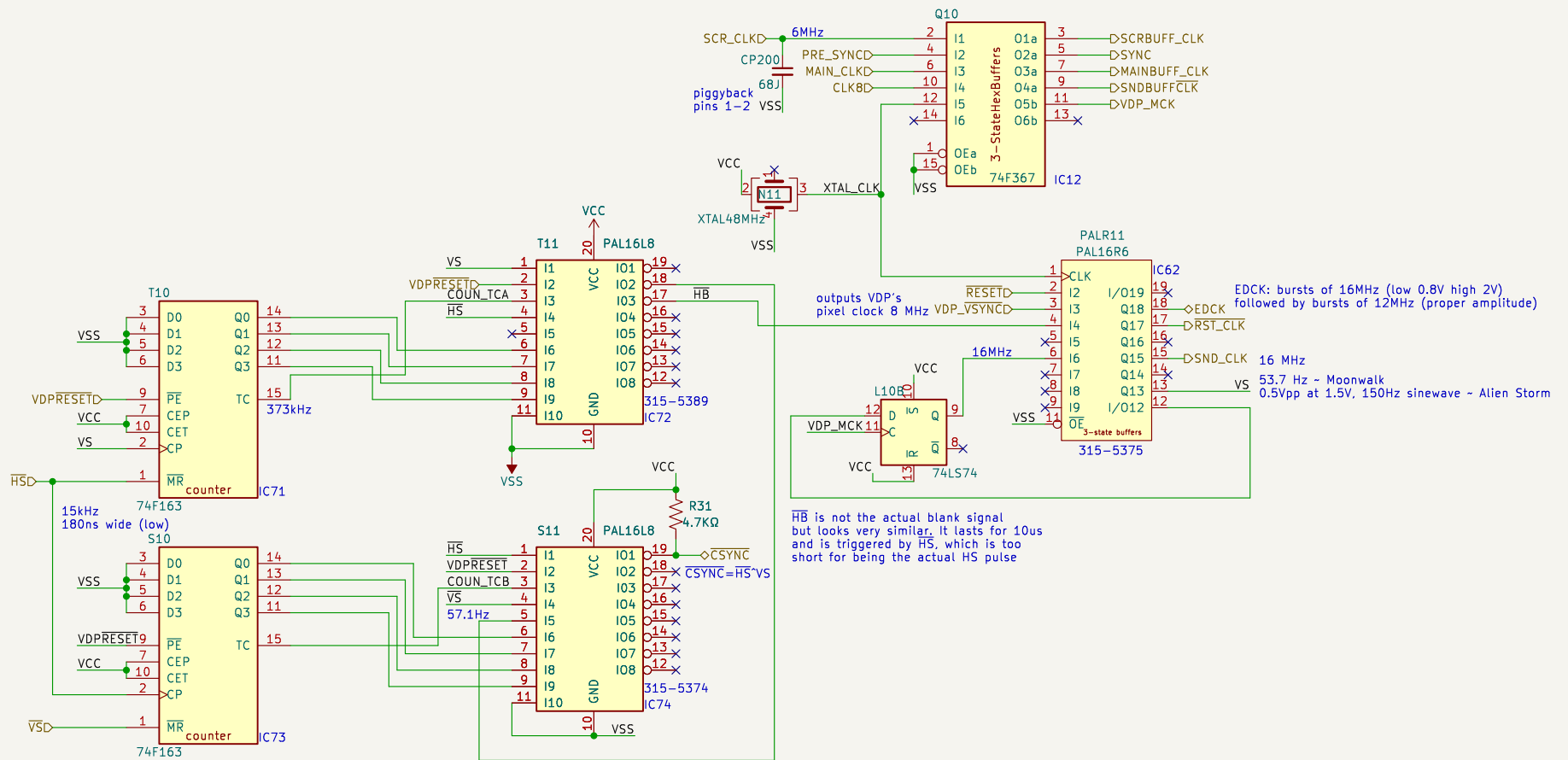
Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	Rev: José Tejada
KiCad E.D.A. 8.0.9		Id: 23/26









VDP pixel clock and csync generation
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Sheet: /Video Sync/
File: buffers.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 26/26

